



Experimental study on the immersion liquid cooling performance of high-power data center servers

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ABSTRACT

Highly dense and integrated data centers face key challenges of realizing efficient cooling and improved energy efficiency. To overcome these challenges, this study experimentally investigated the flow and heat transfer characteristics of single-phase immersion liquid cooling (SPILC) systems. The influence of two opposite coolant flow directions on the SPILC performance was examined. Furthermore, a correlation mechanism between coolant thermal properties and SPILC performance was established, along with a control chart for regulating the working conditions of SPILC systems. The results indicate that compared to a pro-gravity flow, an anti-gravity flow scheme reduces the chip case temperature and thermal resistance by 33.8% and 55.6%, respectively, while decreasing the power usage effectiveness (PUE) by 1.4%. Using coolant with the lowest viscosity reduces the chip case temperature and thermal resistance by 9.3% and 10.5%, respectively, while decreasing the PUE by 0.4%. Moreover, the cooling water temperature has a greater impact on the performance of SPILC systems than the volume flow rate of coolants. Additionally, this paper provides control charts for the cooling water temperature and coolant flow rate to improve the PUE while ensuring the safe operation of SPILC systems, with the highest chip temperature and total electricity consumption as indicators.

1. Introduction

As the new generation of information technology has developed rapidly, society has gradually entered a new era that places high emphasis on digital productivity. To provide the enormous amount of arithmetic support required for the development of our modern information society, data centers inevitably increase in integration, scale, and density [1]. Considering these conditions, data centers will account for 8% of the total global electricity consumption by 2030, and the cooling demand for a single data center rack will rise to 30 kW [2,3]. However, most data centers still use conventional air cooling, which can only address cooling scenarios where the power of a single rack does not exceed 15 kW. Moreover, the cooling energy consumption of data centers accounts for 30%–45% of their total power usage [2,4]. Therefore, the development of efficient and economical cooling technology is an inevitable requirement for the efficient and safe operation of data centers in the future that can also significantly enhance the energy efficiency level and promote the development of related clean energy

implementations [5].

Compared to other methods (i.e., heat pipe [6] and solid-liquid phase change [7,8]), liquid cooling technology has gained widespread attention in data center cooling due to its advantages of strong heat dissipation capability, low noise, and compact structure [9,10]. Depending on whether the coolant is in direct contact with cooled objects, liquid cooling technology is characterized as indirect liquid cooling and direct liquid cooling. Compared with indirect liquid cooling technologies such as liquid cooling panels and micro-channels, immersion liquid cooling technology submerges electronic devices directly in the coolant; thus, the heat transport capacity and variability of electronic devices are further improved. Existing research shows that immersion liquid cooling can be categorized into single-phase immersion liquid cooling (SPILC) [11–13] and two-phase immersion liquid cooling (TPILC) [14–16]. Wang et al. [17] built a pump-driven SPILC system using mineral oil as the coolant. The highest temperature and maximal temperature variability of electronic devices were 39.4% and 74% lower, respectively, than those of conventional air-cooling systems. Kanbur et al. [18]

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quantitatively compared the cooling performances of SPILC and TPILC technologies. The results showed that the COP of the TPILC system was 1.72–1.79 times higher than that of the SPILC system. In general, the SPILC technology offers a more energy-efficient, high-performance, and environmentally friendly solution. Although SPILCs lack cooling capacity compared with TPILCs, their simple structure, easy maintenance, safety, stability, and lower cost make them more suitable for large-scale promotion and commercial application. At present, because most immersion coolants are too expensive, many scholars use simulation methods in the initial exploration stage. Jithin et al. [19] evaluated the cooling capacity of different coolants (i.e., deionized water, mineral oil, and e-fluoride) for SPILC systems. They found that deionized water inhibits the increase in temperature of electronic devices better than the other two coolants. Cheng et al. [20] designed a pump-driven SPILC system with a finned heat sink and investigated its heat transfer performance at different flow rates and heat sink materials. The temperature of the central processing unit (CPU) was negatively correlated with the flow rate of coolants, and the CPU temperature decreased by 16.1% as the coolant flow rate changed from 0.4 to 1.2 m/s. Wen et al. [21] fabricated five new nanofluids based on FC-40 and simulated their application on a SPILC system. The results showed that AI-FC40 exhibited the best heat transport capability because it had the highest thermal conductivity, and the maximum CPU temperature decreased by 6.5% relative to that of FC-40 without added nanoparticles. Muneeshwaran et al. [22] used orthogonal analysis to numerically optimize the heat sink structure of SPILC systems. Their findings indicated that compared to normal heat sinks, heat pipes, and vapor chamber bases can decrease the temperature of electric equipment by 3.3 °C and 4.8 °C, respectively. Compared to numerical studies, relatively few experimental studies on SPILC technology have been conducted, and these few have mainly focused on coolant selection, operating conditions, and heat dissipation enhancement. Luo et al. [23] designed a new silicon carbide-based nanofluid for SPILC systems, and a simulation revealed that at high/low Reynolds number, 3.7 vol%/0.3 vol% nanofluid can enhance the heat transfer effect by 11.7% and 11.4%, respectively, in comparison with traditional mineral oil. Hnayno et al. [24] experimented on the performance of different coolants applied to a SPILC system. The results showed that the cooling ability decreased by approximately 6% when the viscosity of the coolant was changed from 4.6 to 9.8 mPa s. Shrigondekar et al. [25] experimentally investigated the performance of a SPILC system under various operating conditions (i.e., different inlet and outlet positions and circulation flow). The highest temperature and thermal resistance of a SPILC cabinet with a T-type inlet/outlet arrangement decreased by 4.9% and 12.6%, respectively, compared to those of U-type and Z-type arrangements.

This literature review demonstrates that current experimental studies on the application of pump-driven SPILC in data centers have focused mainly on the chip-level (considering only CPUs) [20,26] and module-level (considering only 1U server boards) [27–31] scales. However, research on the system level (i.e., liquid-cooled cabinets composed of two or more server boards) is relatively scarce. In particular, the total heat load of electronic equipment under chip-level and module-level scale studies in the literature is generally between 100 and 300 W, which significantly deviates from the high heat dissipation requirements of servers in practical applications. Therefore, it is necessary to conduct system-level experimental research on pump-driven SPILC systems under high heat loads. In addition, existing studies generally use an anti-gravity flow for the cooling coolant, while Huang et al. [30] showed that the impact of natural convection on the cooling ability of pump-driven SPILC systems cannot be neglected. However, the development of natural convection in SPILC systems under different coolant flow directions (i.e., anti-gravity and pro-gravity flows) will inevitably be affected, thereby influencing SPILC performance. Therefore, it is essential to thoroughly elucidate the influence mechanism of different coolant flow directions on the performance of pump-driven SPILC systems. Moreover, although previous work has been done to compare the

cooling capacity of different coolants used in SPILC systems [24,32], research on the coupling relationship between the coolant thermal properties and the cooling capacity of pump-driven SPILC units is still in the preliminary exploration stage. In particular, the extent of the impact of the different thermal properties of coolants on the thermal transport, fluid flow, and energy efficiency characteristics of the pump-driven SPILC and their interrelationship mechanisms still require further in-depth research. To achieve efficient and reliable operation of immersion liquid-cooled data centers, existing research has investigated the operational characteristics of SPILC systems under different operating conditions. However, most existing studies have primarily focused on the heat transfer characteristics of SPILC, with only a few analyzing the impact of operating conditions on energy consumption levels. Significantly, there has been limited research delving into the extent to which parameters of operating conditions regulate the performance and provide efficient, reliable, and low-carbon operational control strategies for SPILC systems.

To overcome the aforementioned challenges in the SPILC field, this paper reports a visual experimental study on the SPILC in data centers. To achieve this goal, this study establishes a system-level pump-driven SPILC experimental platform that includes multiple server boards. The impact mechanism of the flow direction (pro-gravity and anti-gravity) of the coolant inside a liquid-cooled cabinet on the cooling characteristics of SPILC systems is analyzed. Moreover, the heat transfer characteristics, flow behavior, and energy efficiency of SPILC systems using different coolants are investigated. This study reveals the inherent connection between the SPILC characteristics and coolant thermal properties. Additionally, this study elucidates the regulation mechanism of various operating conditions on the SPILC performance, thus providing theoretical and data-based support for the practical operation of SPILC systems.

2. Experimental methodology

2.1. Experimental system

To analyze the heat transfer, dynamic flow, and energy efficiency characteristics of SPILC systems, the SPILC experimental platform presented in Fig. 1 was constructed in this work. As depicted in Fig. 1(a), the SPILC experimental platform is mainly composed of an experimental test unit, fluid circulation unit, cooling unit, and data acquisition unit. The experimental test unit includes an immersion liquid cabinet, demo servers, a liquid level meter, and a voltage regulator. The demo server is immersed in the electronic fluorinated liquid (EFL) in the immersion liquid cabinet. The voltage regulator can regulate the input power consumption of the demo server, and the liquid level meter is used to ensure that consistent volumes of EFLs are injected. The fluid circulation unit has a gear pump, valves, and pipes. A gear pump is used to regulate the EFL flow rate. The cooling unit mainly includes a plate heat exchanger, tubes, and a chiller. The chiller can regulate the cooling water temperature entering the plate heat exchanger. The data acquisition unit includes temperature transmitters, pressure transmitters, a gear flow rate meter, and a power meter. To obtain the temperature rise and pressure loss of the EFL through the immersion liquid cabinet, temperature/pressure transmitters are arranged at the import and export of the immersion liquid cabinet. Moreover, temperature/pressure transmitters are installed before and after the gear pump to monitor the pressure loss in the internal circulation loop. In addition, the EFL volume flow rate is monitored by a gear flow meter. A photograph of the SPILC experimental platform is given in Fig. 1(b). To minimize the effect of environmental heat loss, the immersion liquid cabinet and the pipelines in the internal and external circulation loops are wrapped with insulating cotton and encapsulated with tin foil. The main experimental devices and their performance parameters are listed in Table 1.

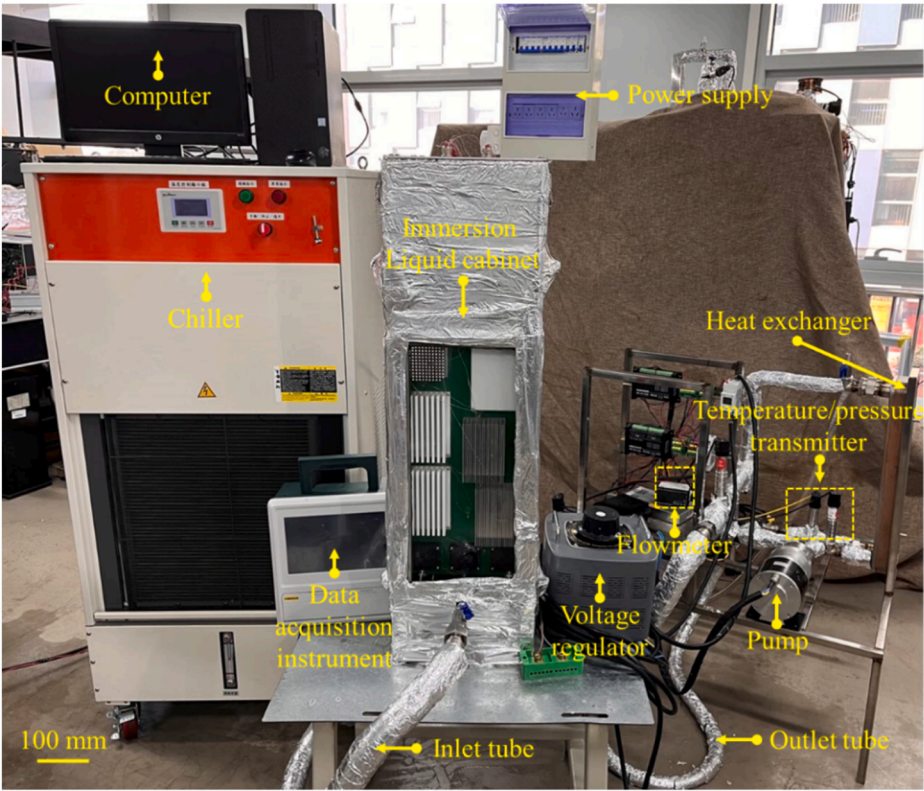
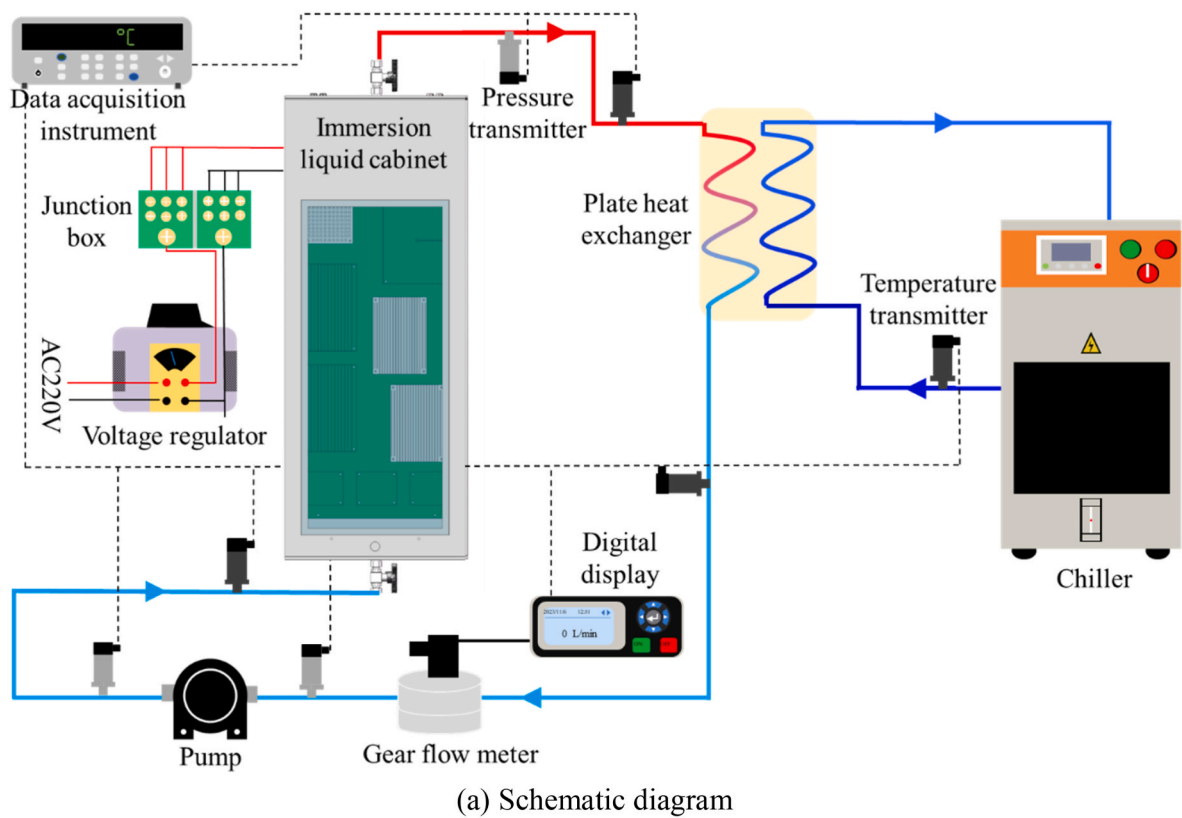


Fig. 1. (a) Schematic and (b) photograph of SPILC experimental platform.

Table 1
Performance parameters of the main devices.

Devices	Brand	Model	Performance parameters
Liquid level meter	NSTRUMENT	HCL300	Range: 0–1 m; Accuracy: ± 0.005 m
Thermocouple	OUCG	TT-T-36-SLE	Temperature: 200–260 °C; Accuracy: ± 0.5 °C
Power meter	Acrel	ADL400	Voltage ≤ 250 V; Current ≤ 80 A; Accuracy: ± 2 W
Gear pump	suofu	NP900	Flow rate ≤ 28 L/min; Temperature ≤ 130 °C; Pressure ≤ 20 bar
Gear flow rate meter	jishen	CX-M12-SS	Range: 0–28 L/min; Accuracy: ± 0.3 L/min
Temperature transmitter	NSTRUMENT	T100	Temperature: 0–150 °C; Accuracy: ± 0.5 °C
Pressure transmitter	NSTRUMENT	HCP100	Pressure: 0–200 kPa; Accuracy: ± 400 Pa
Plate heat exchanger	Alfa Laval	ZL26-20HS	Rated heat transfer power: 3.5 kW
Chiller	RENOVO	FAW-2ABPT	Water temperature: 5–40 °C; Accuracy: ± 1.5 °C

2.2. Immersion liquid cabinet

The immersion liquid cabinet is the core of SPILC systems, and its three-dimensional structural layout is shown in Fig. 2. This assembly drawing shows that the proposed immersion liquid cabinet mainly consists of cabinet shells, the EFL, demo servers, and ancillary accessories. The cabinet shell is welded to aluminum alloy plates with dimensions of $270 \times 136 \times 832$ mm and a thickness of 5 mm. To monitor the internal operating characteristics, a glass window is arranged in the front panel of the immersion liquid cabinet. In this experiment, three different EFLs (company: Zhejiang Juhua; internal types: EFL-1, EFL-2, and EFL-3) are chosen as the immersion coolants to explore the influence of various EFLs on the performance of SPILC systems. The main thermal properties of these three EFLs are listed in Table 2.

Our previous studies demonstrated that natural convection has an effect on the performance of pump-driven SPILC systems [30]. For this reason, two flow directions of the EFL (i.e., pro-gravity and anti-gravity flow directions) are considered in this experiment, as shown in Fig. 2 (b). Notably, the server used in this experiment is a self-designed demo server that mimics the layout of a blade server, and its specific structure is shown in Fig. 2(c). Each demo server consists of a printed circuit board (PCB), demo chips (i.e., the CPU and platform controller hub (PCH)) and their heat sinks, dual-inline-memory-modules (DIMM), a power supply unit (PSU), and hard disk drives (HDDs). Based on our previous research results [30] and a literature review [27,32], the effects of DIMMs, PSUs, and HDDs on immersion cooling performance are not significant. Therefore, to simplify the experiment, this study does not consider the impact of the thermal load on the SPILC system, which is consistent with the literature. For convenience in subsequent discussions, this paper names CPU1 on #1 PCB as CPU1-1, and the other demo chips are named in the same manner. These PCBs are standardized units that are available in the marketplace, the heat sinks of demo chips are machined from the 6061 aluminum alloy, and the fake DIMM, PSU, and HDD are made of nylon (type: PA6). In particular, the demo chip is a heat source module combined with a high-power polyimide electric heating film (thickness: 0.3 mm) that is pasted on the bottom surface of a metal block. Based on engineering simulation experience, the equivalent thermal conductivity of components such as CPUs is $10\text{--}40 \text{ W m}^{-1} \text{ K}^{-1}$. Therefore, stainless steel is chosen as the metal block for simulating the chip herein. Notably, the side of the heating film that does not come into contact with the stainless steel block is insulation. Moreover, the heating film is located between the demo chip and the PCB. The power of demo chips is adjusted by adjusting the voltage regulator and monitored by a power meter. To reduce the contact thermal resistance, thermal conductive silicone grease No. TC-5888 ($\lambda = 5.2 \text{ W m}^{-1} \text{ K}^{-1}$) is added

between the surfaces of demo chips and its heat sink base. Table 3 gives the key information of components on the demo servers. For the EFL region, T-type thermocouples are also included to monitor the temperature uniformity of the EFL inside the immersion liquid cabinet. As shown in Fig. 2 (d), 14 temperature measurement points are arranged on each PCB, on which thermocouples are adhered 15 mm and 40 mm from the PCB surface. The selection of thermocouple positions is primarily for a more accurate assessment of the temperature uniformity in the EFL and to facilitate calculating the local Nusselt number near the CPU. Moreover, T-type thermocouples (accuracy: ± 0.5 °C) are adhered to the center of the surfaces of demo chips and heat sink fin to monitor the temperature response characteristics.

2.3. Data reduction and uncertainty analysis

2.3.1. Data processing

To quantitatively analyze the flow and thermal characteristics of SPILC systems, all the measuring points are monitored and recorded in real-time during the experimental process. Moreover, several evaluation indices are applied based on experimental data for the performance assessment of SPILC systems. For the heat transfer characteristics, the case temperature of demo chips, the equivalent thermal resistance of demo chips, and the temperature variability index of EFLs are used as the evaluation indices. To evaluate the dynamic flow characteristics of SPILC systems, the pressure drop of EFLs between the import and export of the immersion liquid cabinet, the pressure drop of the internal circulation loop, and the localized Nusselt number are used as the evaluation indices. Moreover, the power usage effectiveness (PUE), which is typically employed in data centers, is used as an index for evaluating energy efficiency characteristics.

(1) Equivalent thermal resistance

The equivalent thermal resistance is introduced to evaluate the heat dissipation capability of coolants, which is defined as the thermal resistance between the case of a demo chip and the inlet coolant.

$$R_{th} = \frac{T_d - T_{in}}{P_1} \quad (1)$$

where T_d and T_{in} denote the case temperature of a demo chip and the temperature of the inlet coolant, respectively, and P_1 denotes the power of the demo chip.

(2) Temperature variability index

To evaluate the temperature distribution characteristics of EFLs inside an immersion liquid cabinet, this paper defines the standard deviation of all temperature measurement points in EFL regions as the temperature variability index, which can be calculated by

$$\sigma_T = \sqrt{\frac{\sum_{i=1}^n \left(T_i - \frac{\sum_{i=1}^n T_i}{n} \right)^2}{n}} \quad (2)$$

where T_i denotes the temperature measurement points in the EFL region, and n is the number of measurement points. Notably, the higher the temperature variability index is, the worse the temperature uniformity in the EFL region.

(3) Localized Nusselt number

The localized Nusselt number is usually used to characterize the heat transfer capacity in a specific flow region. In this experiment, the localized Nusselt number refers to the Nusselt number in the vicinity of

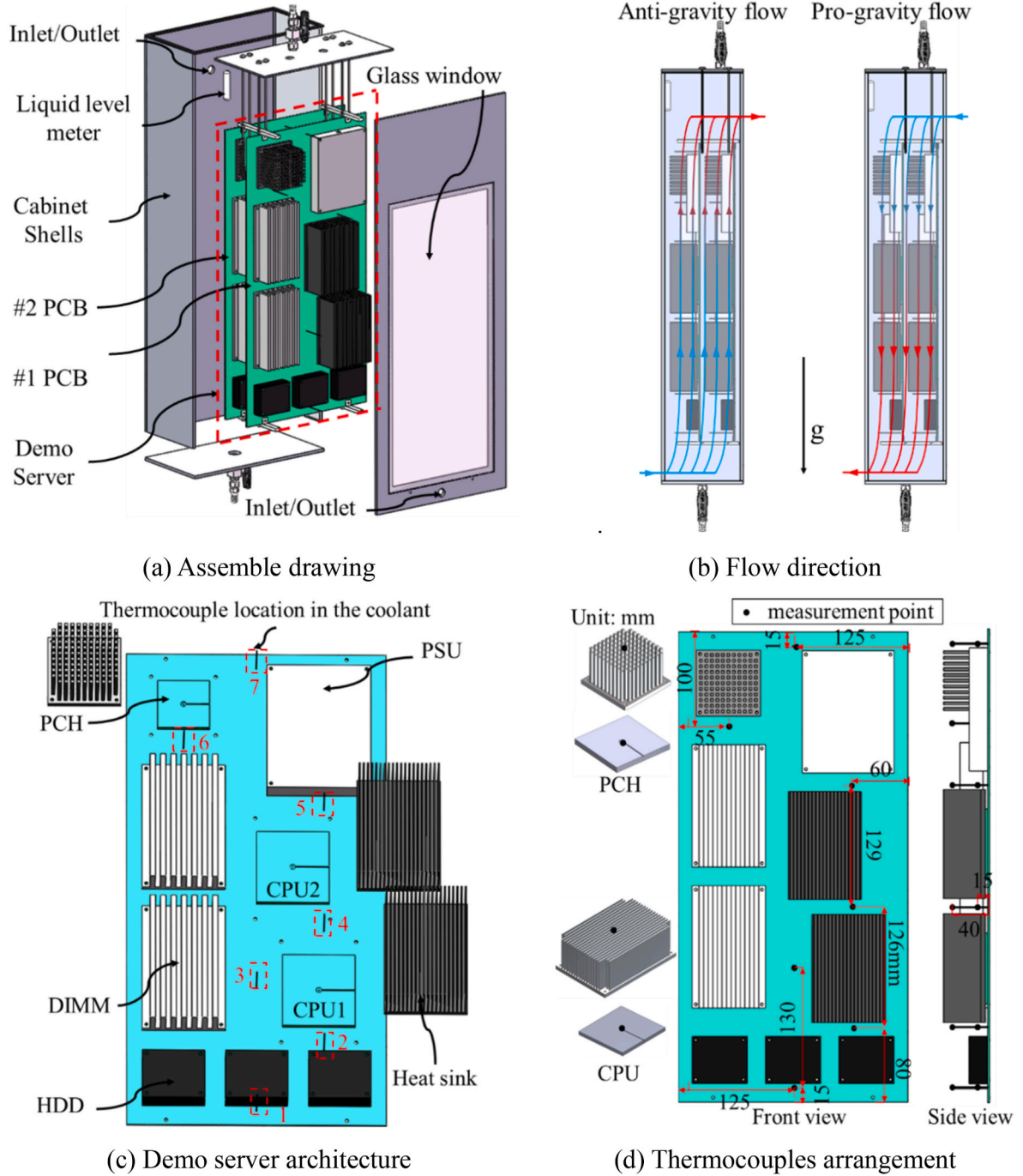


Fig. 2. Structural description of the immersion liquid cabinet.

the CPU, which can be calculated by

$$Nu = \frac{P_l l}{A \Delta T \lambda} \quad (3)$$

where l denotes the feature length of the CPU heat sink, A denotes the surface area of a CPU heat sink, λ denotes the thermal conductivity of the EFL, and ΔT denotes the average temperature difference between the heat sink and the nearby EFL. Notably, the average temperature of EFLs is the average of the EFL temperatures before and after the CPU. Considering that the thermal conductivity of heat sinks is much higher than that of EFLs, this paper approximately considers the measurement temperature on the heat sink as the average temperature of the heat sink.

(4) Pressure drop

To determine the energy loss of the EFL, this paper introduces the pressure drop of the immersion liquid cabinet and internal circuit loop as the evaluation indices of the dynamic flow characteristics, which can be calculated by

$$\Delta p_{\text{cabinet}} = p_{\text{cabinet,in}} - p_{\text{cabinet,out}} \quad (4)$$

$$\Delta p_{\text{system}} = p_{\text{pump,out}} - p_{\text{pump,in}} \quad (5)$$

where Δp is the pressure drop, $p_{\text{cabinet,in}}$, $p_{\text{cabinet,out}}$ respectively denotes the pressure of cabinet inlet and outlet, $p_{\text{pump,in}}$, $p_{\text{pump,out}}$ respectively denotes the inlet and outlet pressure of the gear pump in the internal

Table 2
Main thermal properties of EFLs (1 bar).

Thermal properties	Temperature (± 3 °C)	EFL-1	EFL-2	EFL-3
Boiling point temperature (T)/°C		177.3	286.2	172.5
Density (ρ)/kg·m ⁻³		1889	1894	1883
Specific heat capacity (c_p)/ J·kg ⁻¹ ·K ⁻¹		1165	1338	1219
Thermal conductivity (λ)/W·m ⁻¹ ·K ⁻¹	20	0.062	0.062	0.057
	40	0.068	0.064	0.059
	60	0.072	0.070	0.064
Dynamic viscosity (μ)/mPa·s	20	6.31	3.17	2.79
	40	2.77	2.15	1.49
	60	1.72	1.73	1.45
Price/\$·kg ⁻¹		41.7–55.6	25–27.8	

Table 3
Key parameters of main devices on each demo server.

Components	Geometric dimensions (mm)	Power (W)	Quantity
PCB	500 × 250 × 2		1
CPU	75 × 70 × 4.8	0~350	2
PCH	50 × 50 × 4.8	0~150	1
DIMM	130 × 80 × 30		4
PSU	130 × 100 × 20		2
HDD	50 × 60 × 20		6
Heat sink of CPU1/ CPU2	Base: 115 × 80 × 5 Fin: 115 × 1 × 40; Fin pitch: 3		1 20
Heat sink of PCH	Base: 70 × 70 × 5 Fin: 3 × 3 × 40; Fin pitch: 3		1 107

circulation circuit.

(5) PUE

PUE is an internationally accepted measure of power usage efficiency in data centers. It is defined here as the ratio of the total power consumption to the power consumption of demo chips.

$$PUE = \frac{P_{IT} + P_{pump} + P_{chiller}}{P_{IT}} \quad (6)$$

where the subscripts *IT*, *pump*, and *chiller* denote the demo chip, gear pump and chiller, respectively.

2.3.2. Uncertainty analysis

In this experiment, the directly measured parameters include the temperature T , pressure p , volume flow rate Q_v , and power P . The indirectly measured parameters include the equivalent thermal resistance R_{th} , temperature variability index σ_T , localized Nusselt number Nu , pressure drop Δp , and PUE. According to the error transfer theory of Moffat [33], the relative uncertainties of directly measured parameters (δ_d) and indirectly measured parameters (δ_i) can be calculated by the following equation:

$$\delta_d = \frac{|\sigma_p|}{d_{min}} \quad (7)$$

$$\delta_i = \sqrt{\left[\left(\frac{\partial F}{\partial p_1} \right)^2 \sigma_{p_1}^2 + \left(\frac{\partial F}{\partial p_2} \right)^2 \sigma_{p_2}^2 + \dots + \left(\frac{\partial F}{\partial p_n} \right)^2 \sigma_{p_n}^2 \right]} / F_{min} \quad (8)$$

where, σ_p denotes the measurement accuracy of directly measured parameters, d_{min} represents the minimum value of directly measured parameters, and F represents the functional relationship between the indirectly and directly measured parameters. Table 4 lists the relative uncertainties of important parameters.

Table 4
Relative uncertainties of directly and indirectly measured parameters.

Parameter	Symbol	Unit	Relative uncertainty
Temperature	T	°C	3.3%
Pressure	p	kPa	2.5%
Volume flow rate	Q_v	L/min	13.35%
Power	P	W	2%
Thermal resistance	R_{th}	°C/W	4.81%
Temperature variability index	σ_T	°C	11.15%
Localized Nusselt number	Nu		6.29%
Pressure drop	Δp	kPa	3.53%
PUE			1.74%
Power consumption	Q	kW·h	5.21%

3. Results and discussion

3.1. Reproducibility verification of experimental data

To ensure the reliability of the experimental data reported in this paper, a repeatability verification of the SPILC experimental platform is conducted prior to the formal experiment. In the repeatability verification, the total power of demo chips in the immersion liquid cabinet is 1.75 kW. The type of coolant is EFL-1, and the coolant is in an anti-gravity flow direction. The EFL volume flow rate is 2 L/min, the cooling water temperature is 15 °C, and the inlet temperature of EFLs is 18 °C. Fig. 3 presents the measured case temperature (i.e., T_d) for different demo chips in three repeated experiments under the same operating conditions. The data clearly show that the consistency of the measured results is good among the three different batches of experiments. According to the statistical analysis, the maximum absolute deviation of the case temperature for the same demo chip in different batches of experiments is only 0.9 °C, and the relative maximum deviation is 1.47%, both of which are within the allowable range of experimental error. This result verifies that the experiment is repeatable and the measured data are both accurate and reliable.

Fig. 3 shows that the case temperatures of CPU1 and CPU2 on #2 PCB are lower than their respective values on #1 PCB. This occurs because the coolant enters from the lower part of the front side of the immersion liquid cabinet, resulting in a higher distribution of coolant flow to #2 PCB, which is located at the back of the immersion liquid cabinet. Thus, the heat transfer for the CPUs located in the lower and middle parts of the #2 PCB is improved. Additionally, CPU2 experiences a significant increase in the case temperature compared to CPU1 due to

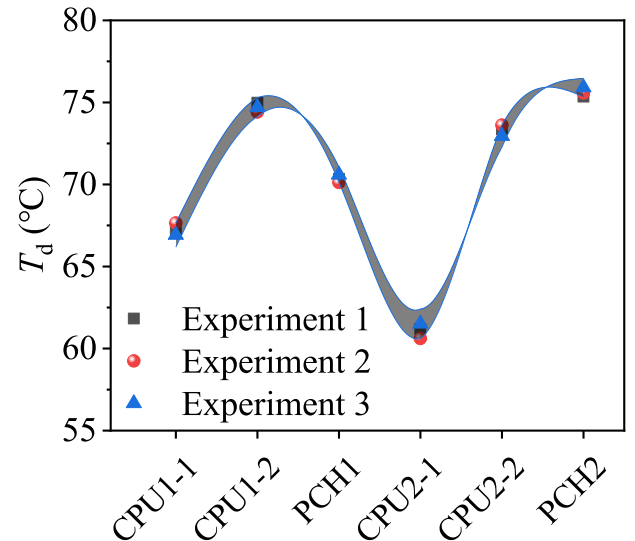


Fig. 3. Experimental reproducibility verification.

the thermal barrier effect from the upstream CPU1. Notably, because #2 PCB is closer to the outlet, it is more affected by the thermal barrier from upstream, leading to more severe deterioration in heat dissipation for the PCH in the upper part of #2 PCB compared to #1 PCB. In conclusion, the position of demo chips on the PCB and their placement in the immersion liquid cabinet both have a significant impact on their immersion cooling performance. To guarantee the efficient function of the server, high-power devices should be arranged upstream in the main-stream direction in the immersion liquid cabinet.

3.2. Performance comparison of SPILC systems under two flow directions

To further explore the effects of coolant flow directions (i.e., pro-gravity and anti-gravity flow) on the thermal transport, flow behavior, and energy efficiency characteristics of SPILC systems, this study compares the experimental results of two flow modes under the same operating conditions (i.e., a total chip power of 1.75 kW in the immersion liquid cabinet, an EFL-1 coolant volume flow rate of 2 L/min, a cooling water temperature of 15 °C, and the inlet temperature of EFL-1 is 18 °C). Fig. 4 provides the case temperature and thermal resistance of demo chips in an immersion liquid cabinet under two coolant flow directions. The data show that, compared to the coolant flowing in the pro-gravity direction, the coolant flowing in an anti-gravity direction can result in lower case temperatures and thermal resistances of demo chips. Statistical analysis reveals that the case temperature and thermal resistance of CPU1-2 in the immersion liquid cabinet with coolant flowing in an anti-gravity direction decrease by 33.84% and 55.55%, respectively, compared to cases where the coolant flows in the pro-gravity direction. This phenomenon is attributed to the presence of natural convection. When the coolant flows against gravity, it enhances the mixing and circulation of the EFL inside the SPILC system, thus improving the immersion cooling performance of the demo chips. In anti-gravity flow, the case temperature of CPU2-2 is 12.16 °C higher than that of CPU2-1 due to the influence of the thermal cascade effect. However, in pro-gravity flow, the case temperature of CPU2-2 is only 2.66 °C higher than that of CPU2-1. This phenomenon indicates that pro-gravity flow can effectively alleviate the thermal cascade effect of CPUs and improve heat transfer uniformity. Interestingly, under the pro-gravity flow direction, the temperature trends of demo chips on the two PCBs are very similar to those under the anti-gravity flow. From Fig. 4(a), it can be observed that the case temperatures of CPU2-1 and CPU2-2 on #2 PCB are lower under two different coolant flow directions compared to those of CPU1-1 and CPU1-2 on #1 PCB. Combining this with Fig. 5(a), it can be noted that the temperature differences of EFLs at the same measuring points on the two PCBs are relatively small under different coolant flow directions.

Therefore, the above phenomenon is attributed to the fact that the coolant volumetric flow rate passing through CPU2-1 and CPU2-2 on #2 PCB is larger than that passing through CPU1-1 and CPU1-2 on #1 PCB. This indicates that the configuration of the immersion liquid cabinet leads to unequal coolant distribution on the two PCBs.

Fig. 5 compares of the temperature distribution and temperature uniformity of coolants in the immersion liquid cabinet under the two flow modes. Fig. 5(a) shows that the coolant in the immersion liquid cabinet with coolant flowing in the pro-gravity direction exhibits better temperature uniformity and higher average temperature than the corresponding case with coolant flowing in an anti-gravity direction. By comparing Figs. 5(a) and Fig. 4(a), it can be observed that the EFL temperature difference at measurement point #2 on the front side of CPU-1 under pro-gravity and anti-gravity flows is greater than the corresponding value at measurement point #4 on the front side of CPU-2, resulting in a significantly larger temperature difference for CPU-1 under different coolant flow directions compared to CPU-2. Furthermore, along the flow direction of EFLs, the maximum temperature difference of EFLs in pro-gravity flow is only 25.2%–31.2% of that in anti-gravity flow. This indicates that the natural convection strength is significantly weakened in immersion cooling cabinets under pro-gravity flow. Additionally, it can be observed that in anti-gravity flow, the EFL maximum temperature occurs at the outlet of the PCB (top region). In contrast, in pro-gravity flow, the EFL maximum temperature appears at the rear side of CPU2 on the PCB (upper middle region). This phenomenon suggests that the development range of natural convection within immersion liquid cooling cabinets also varies under the two different coolant flow directions. The temperature variation index shown in Fig. 5 (b) also indicates that the temperature variation index of coolants in the immersion liquid cabinet with coolant flowing against gravity is 4.25 times higher than that of the corresponding case with coolant flowing in the same direction as gravity. This phenomenon is attributed to the fact that when the coolant flows in an anti-gravity direction, it enhances the strength of natural convection, improving the thermal transport performance and exacerbating the temperature stratification of coolants in the immersion liquid cabinet. Conversely, when the coolant in the immersion liquid cabinet flows in a pro-gravity direction, the natural convection of the coolant is suppressed, leading to deteriorated heat transfer performance and increased temperatures of demo chips. Interestingly, it is precisely because the development of natural convection is limited that it improves the temperature uniformity of coolants in the immersion liquid cabinet. Therefore, the presence of natural convection significantly impacts the thermal characteristics of SPILC systems and should not be ignored.

Considering the heat convective characteristics of SPILC systems

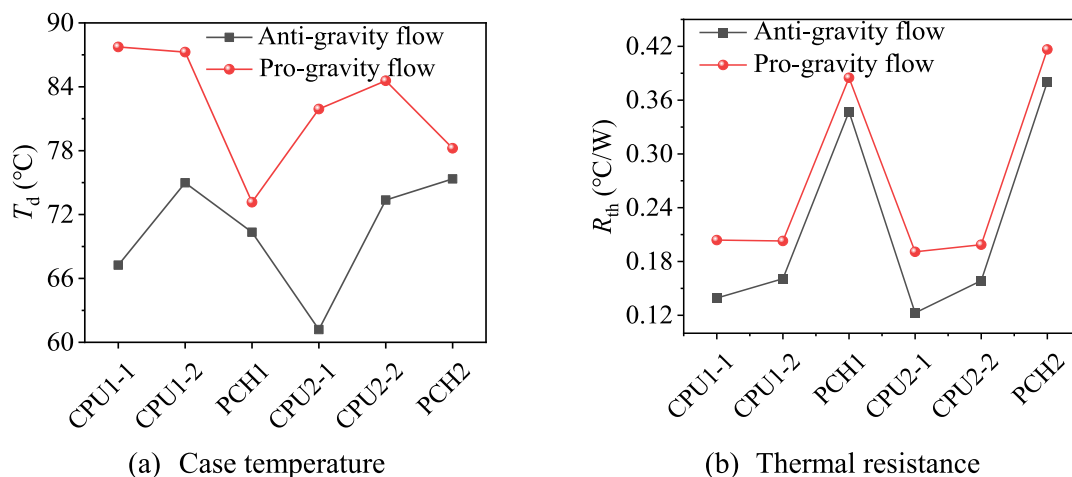


Fig. 4. Heat transfer characteristics of components under two flow directions.

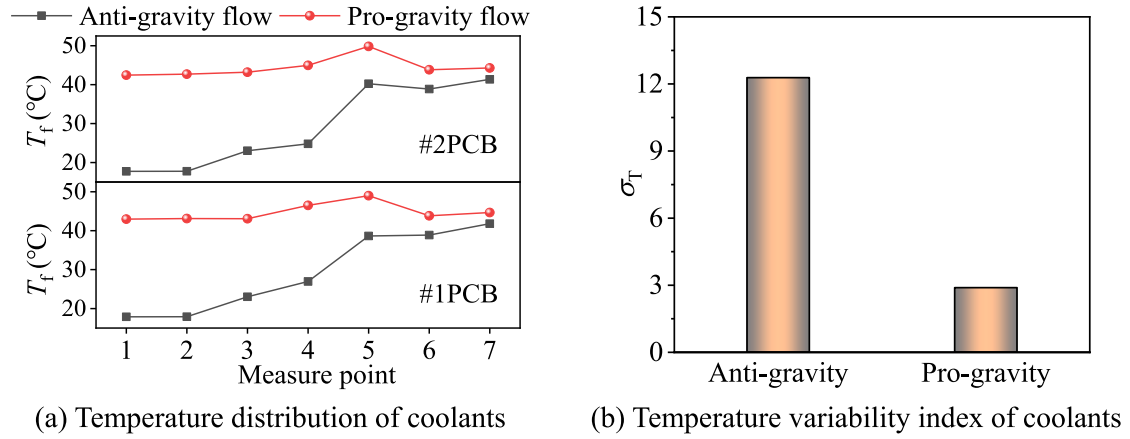


Fig. 5. Temperature response in the fluid domain.

with different coolant flow directions, Fig. 6 presents the pressure drop across the SPILC system and the immersion liquid cabinet, as well as the local Nusselt numbers for the CPU, under two flow directions. Fig. 6(a) shows that when the coolant flows in the pro-gravity direction, the flow resistance of the entire SPILC system decreases to some extent. The pressure drop across the cabinet and the system decreases by 3.82% and 9.78%, respectively, compared to that of the coolant flowing in an anti-gravity direction. Fig. 6(b) shows the local Nusselt numbers for CPU2 under the two different coolant flow directions. As shown, the local Nusselt numbers for CPU1-2 and CPU2-2 in the immersion liquid cabinet with coolant flowing in an anti-gravity direction increase by 121.19% and 157.78%, respectively, compared to the case where the coolant flows in the pro-gravity direction. Therefore, the convective heat transfer in the SPILC system is superior when the coolant flows in an anti-gravity direction. This occurs because when the coolant flows against gravity, it can enhance the convective heat transfer to some extent. Conversely, when the coolant flows in the pro-gravity direction, the convective heat transfer intensity in immersion liquid cabinets is weakened.

To evaluate the energy efficiency of an SPILC system with different coolant flow directions, Fig. 7 quantitatively analyses the pump power consumption and PUE of SPILC systems when the maximum CPU temperature reaches 75 °C. Fig. 7(a) shows that when the coolant in the immersion liquid cabinet flows in the pro-gravity direction, the corresponding pump power consumption of the SPILC system needs to be 2.55 times greater than that of the coolant flowing in the anti-gravity direction to maintain a consistent maximum CPU temperature.

Therefore, the immersion cooling performance in an immersion liquid cabinet is weaker when the coolant flows in the pro-gravity direction than in a SPILC system where the coolant flows in the anti-gravity direction. Fig. 7(b) shows a quantitative analysis of the PUE of the SPILC system under the two flow directions. As observed from the data, compared to the case where the coolant flows in the pro-gravity direction, the PUE of the SPILC system decreases by 1.4% when the coolant flows in the anti-gravity direction. Therefore, the SPILC system with an anti-gravity flow has a higher energy efficiency, which is beneficial for reducing the overall energy consumption of data centers.

3.3. Performance comparison of SPILC systems using three coolants

To clarify the intrinsic relationship between the EFL thermal properties and the SPILC capacity, this study conducted experimental research on the impact of three EFLs on the immersion cooling characteristics. During these experiments, the total power of the demo chips in the immersion liquid cabinet was 1.75 kW. The coolant flows in an anti-gravity direction, with a volume flow rate of 2 L/min, the cooling water temperature is 15 °C, and the inlet temperature of the EFLs is 18 °C. Fig. 8 presents the chip case temperature and thermal resistance of the demo chips inside the immersion liquid cabinet using different coolants. As depicted in the figure, the chip case temperature and thermal resistance of the demo chips are lower when using EFL-3 as the coolant than when using EFL-2 or EFL-1. Furthermore, because PCB #2 is in the main flow region, the response of the chip case temperature on PCB #2 is more sensitive to different EFLs. According to the data analysis,

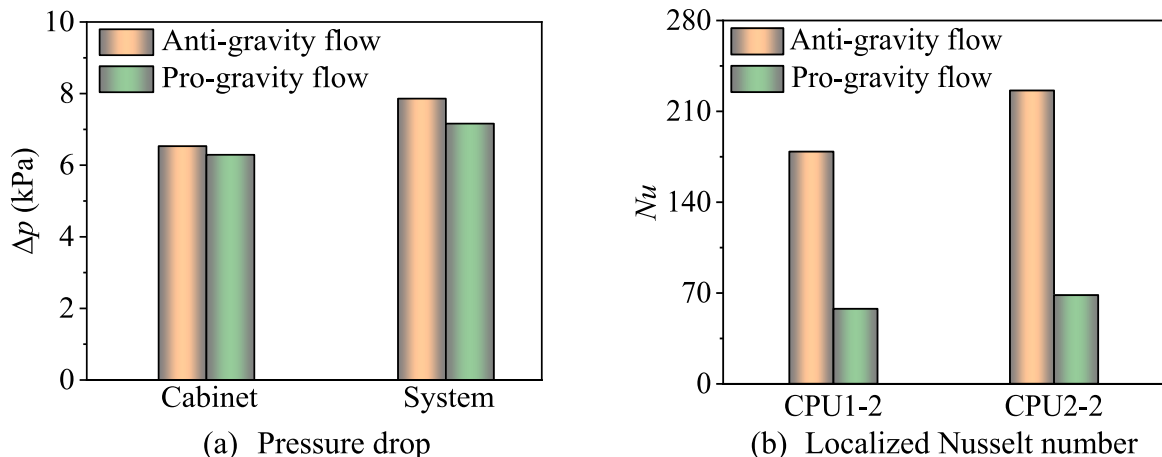


Fig. 6. Heat convection performance of SPILC systems under two flow directions.

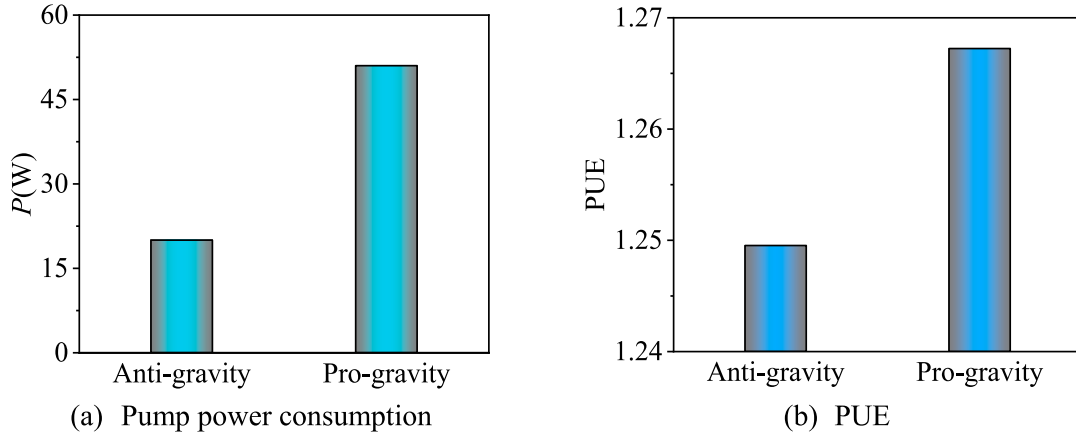


Fig. 7. Analysis of energy efficiency of SPILC systems under two flow directions.

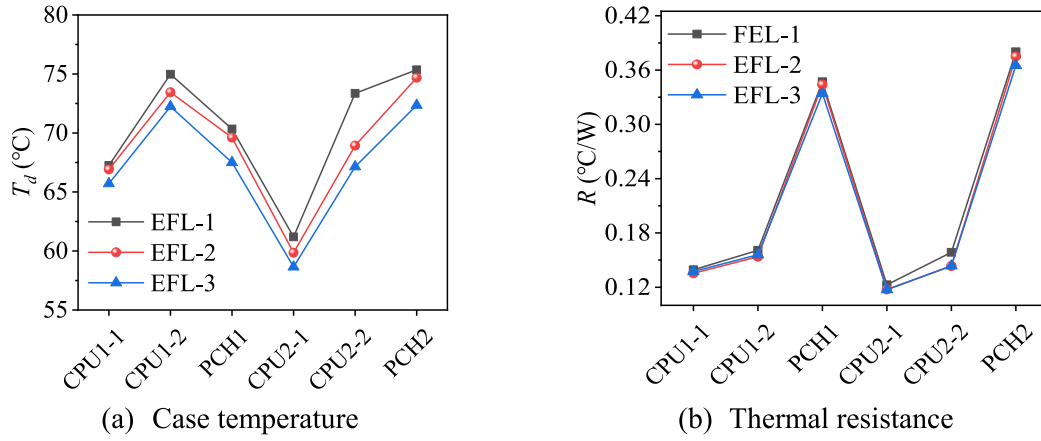


Fig. 8. Thermal transfer characteristics of components with different coolants.

compared to those of EFL-1 and EFL-2, the temperature of CPU2-2 decreased by 9.26% and 2.67%, respectively, when EFL-3 is used as the coolant, and the corresponding thermal resistance decreased by 10.45% and 1.28%, respectively. A detailed comparison of the thermal properties of these three EFLs indicates that the heat conductivity and dynamic viscosity of EFL-1 and EFL-2 are greater than those of EFL-3, while the specific heat capacity of EFL-3 is less than that of EFL-2 and greater than that of EFL-1. Although a larger heat conductivity enhances the thermal transport capability and a higher specific heat capacity

improves the thermal dissipation ability per unit mass of EFLs, the increased dynamic viscosity leads to a thickened boundary layer and deteriorated heat transfer. Therefore, EFL-3 demonstrates the optimal immersion cooling performance due to its lower dynamic viscosity. This phenomenon indicates that the heat transfer enhancement gained from the increase in thermal conductivity and specific heat capacity is not sufficient to counterbalance the heat transfer deterioration caused by the increase in dynamic viscosity.

Fig. 9 further analyzes the temperature distribution and temperature

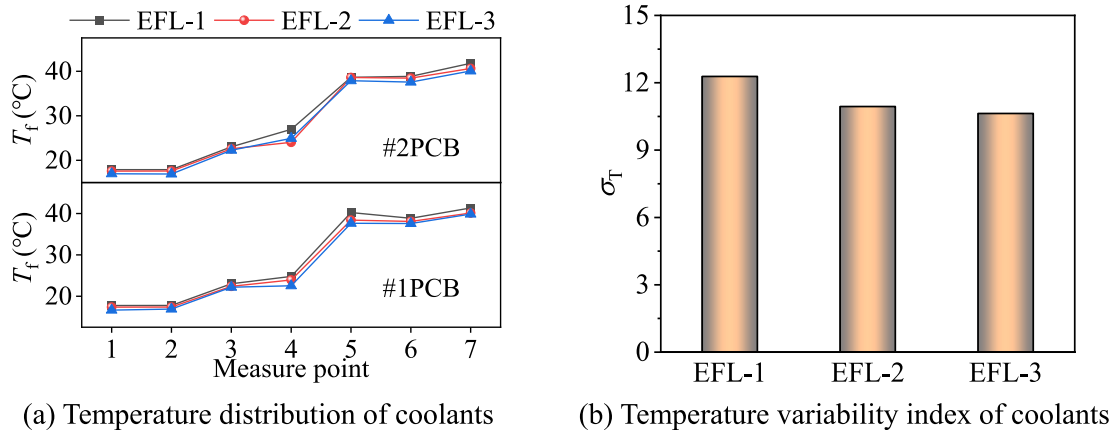


Fig. 9. Temperature response in the fluid domain.

uniformity of different EFLs in the immersion liquid cabinets. Fig. 9(a) depicts that the coolant temperature differences at the same measurement point are relatively small when different EFLs are used in the immersion liquid cabinet. However, closer observation reveals that the temperatures at each measurement point are slightly lower when the coolant is EFL-2 or EFL-3 compared to the case of EFL-1. This occurs because EFL-1 has a higher dynamic viscosity than EFL-2 and EFL-3, while its thermal conductivity is similar to that of the other two EFLs. Therefore, the thicker heat transfer boundary layer caused by the higher dynamic viscosity of EFL-1 results in a worse heat transfer capability than that of the other two EFLs. Additionally, although the dynamic viscosity of EFL-3 is slightly lower than EFL-2, its specific heat capacity is also lower. Therefore, the temperature distribution characteristics inside the immersion liquid cabinets are similar when EFL-2 and EFL-3 are used. Interestingly, the temperature differences at measurement points in the low-temperature region among different EFLs are slightly smaller than the corresponding values at high temperatures. This phenomenon can be explained by the competitive mechanism between heat transfer enhancement through thermal conductivity and heat transfer suppression through dynamic viscosity at different temperatures. In this experiment, the low-temperature region is located upstream of the coolant, with a thinner boundary layer, resulting in a more significant influence of thermal conductivity on heat transfer. As the coolant's downstream temperature increases, the impact of dynamic viscosity on heat transfer gradually increases, leading to larger temperature differences among measurement points under different EFLs. Therefore, it is evident that the dynamic viscosity of EFL at high temperatures has a more significant impact on its SPILC performance. Fig. 9(b) further analyzes the temperature uniformity in the immersion liquid cabinets using different EFLs. Compared to those in EFL-1 and EFL-2, the temperature variability indices in the immersion liquid cabinet using EFL-3 are 15.54% and 2.92% lower, respectively. This result further confirms that a decrease in the dynamic viscosity of EFLs not only enhances the immersion cooling performance but also improves the temperature uniformity of immersion liquid cabinets.

Fig. 10 analyzes the convective characteristics of the SPILC system using three different EFLs. Fig. 10(a) shows the pressure drop of the immersion liquid cabinet and the entire system when different EFLs are used in the SPILC system. Compared to those of EFL-1 and EFL-2, the pressure drop of the immersion liquid cabinet using EFL-3 as the coolant is reduced by 4.48% and 1.28%, respectively, while the corresponding system pressure drop decreases by 6.36% and 2.31%, respectively. This is mainly because the thermal conductivities of these three EFLs are similar, while the dynamic viscosity of EFL-3 is smaller than that of EFL-1 and EFL-2. An increase in the dynamic viscosity leads to a thicker boundary layer on the demo chip surfaces, resulting in increased flow resistance losses. Thus, EFL-3, which has the lowest dynamic viscosity,

exhibits the best flow resistance characteristics. Fig. 10(b) shows that compared to EFL-1 and EFL-2, using EFL-3 as the coolant increases the local Nusselt number of CPU1-2 in the immersion liquid cabinet by 4.06% and 3.21%, respectively, and the local Nusselt number of CPU2-2 also increases by 3.43% and 2.21% respectively. Therefore, a lower dynamic viscosity helps weaken the boundary layer on the demo chip surface, enhancing the immersion cooling effect between the coolant and demo chips.

To research the effect of various EFLs on energy consumption, Fig. 11 quantitatively analyzes the pump power consumption and PUE of the SPILC system using three different coolants. To ensure a fair comparison, the pump speed is adjusted to achieve a maximum CPU case temperature of 72.5 °C for all three EFLs in the immersion liquid cabinets. Fig. 11(a) shows that only when the pump power consumption increases by 47.06% and 17.65%, respectively, can the CPU case temperatures for EFL-1 and EFL-2 match that of EFL-3. This result further indicates that the immersion cooling performance is superior when using EFL-3 compared to the other two EFLs. Fig. 11(b) quantitatively evaluates the PUE of the SPILC system using the three EFLs. Compared to those of the cases using EFL-1 and EFL-2 as coolants, the PUE of the SPILC system using EFL-3 as the coolant decreases by 0.4% and 0.2%, respectively. Therefore, using EFL-3 as the coolant, which has a lower dynamic viscosity, not only enhances the SPILC performance but also decreases the energy consumption of the SPILC system.

3.4. Factors influencing the SPILC performance

During the actual operation of the SPILC system, both the cooling water temperature and the circulating flow rate of the coolant inside the immersion liquid cabinet may vary. Therefore, it is necessary to study the heat transfer and flow characteristics of the SPILC system under different operating conditions. In this case, EFL-3, which has the best immersion cooling performance, is chosen as the coolant, and the coolant flows in an anti-gravity direction. Furthermore, during the experiment, the flow rate of the coolant in the internal circulation loop is varied between 2 and 8 L/min, and the cooling water temperature is varied between 15 and 30 °C.

3.4.1. Effect of cooling water temperature

To investigate the impact of cooling water temperatures on the operating characteristics of SPILC systems, the current work quantitatively assesses the temperature response, flow characteristics, and heat convection capability in an immersion liquid cabinet. During the experiment, the total power of demo chips in the immersion liquid cabinet is 1.75 kW, and the EFL-3 flow rate is maintained at 2 L/min. Fig. 12 presents the temperature response of demo chips and EFL inside the immersion liquid cabinet under different cooling water

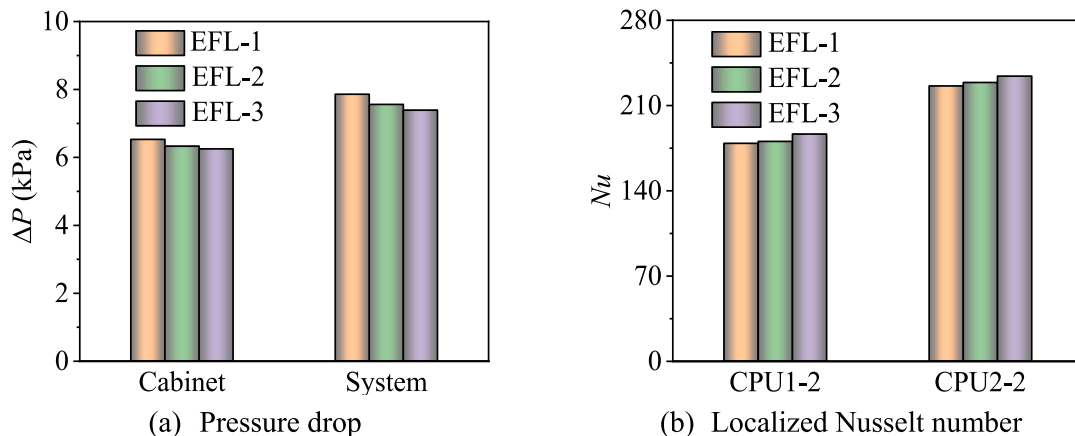


Fig. 10. Heat convection performance of SPILC systems with different coolants.

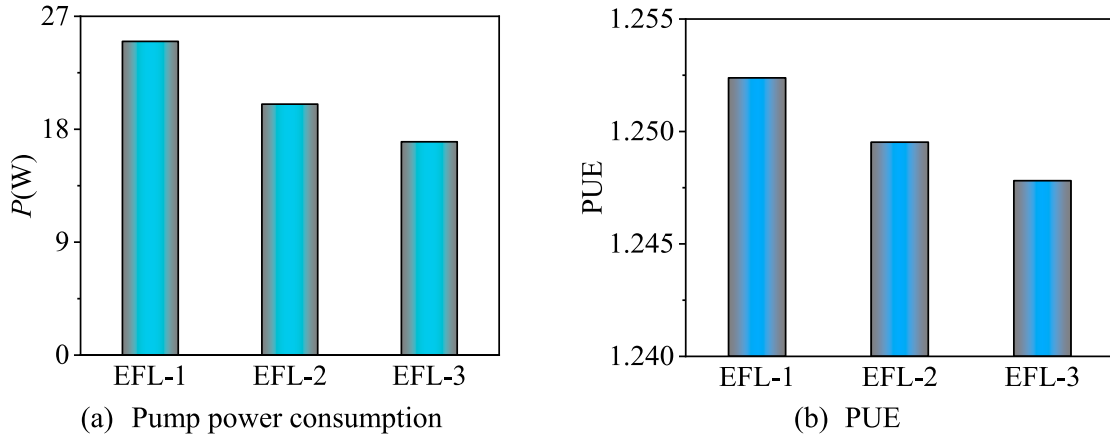


Fig. 11. Analysis of the energy efficiency of SPILC systems with different coolants.

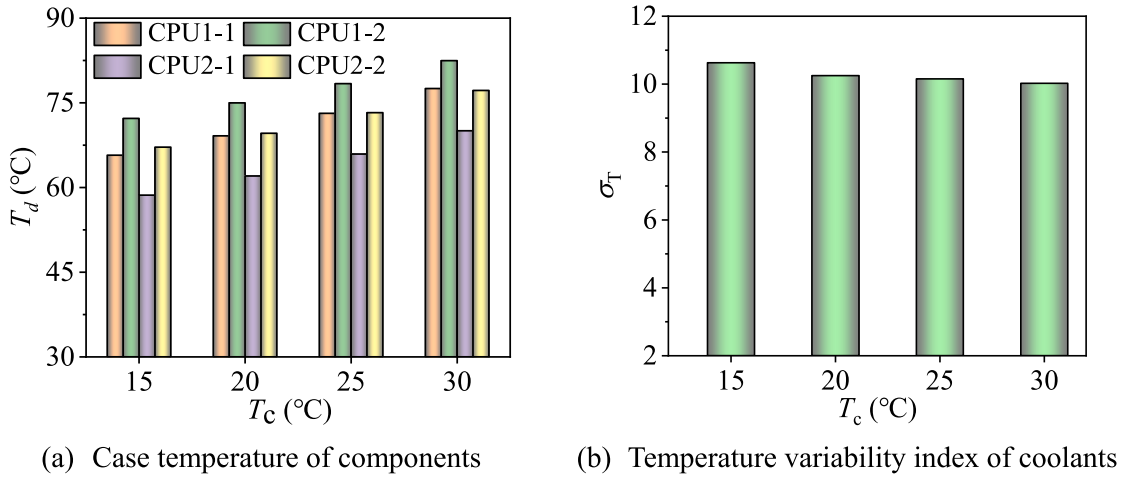


Fig. 12. Effect of water temperature on the temperature response.

temperatures. Fig. 12(a) clearly shows that the case temperature of CPUs increases with the rise in cooling water temperature. Through quantitative analysis, it is concluded that as the cooling water temperature changes from 15 °C to 20, 25, and 30 °C, the case temperature of CPU1-2 rises by 2.75 °C, 6.16 °C, and 10.23 °C, respectively. The increase in the CPU case temperature is less than that in the cooling water temperature, and this discrepancy decreases with increasing cooling water

temperature. This phenomenon is attributed to the increased temperature of the cooling water, leading to a continuous decrease in the dynamic viscosity of EFLs, thinning the boundary layer on the surface of demo chips, thereby intensifying the heat convection between the EFL and demo chips to some extent. However, as the cooling water temperature increases, the rate of change in the EFL viscosity decreases gradually, leading to a gradual decrease in the heat transfer

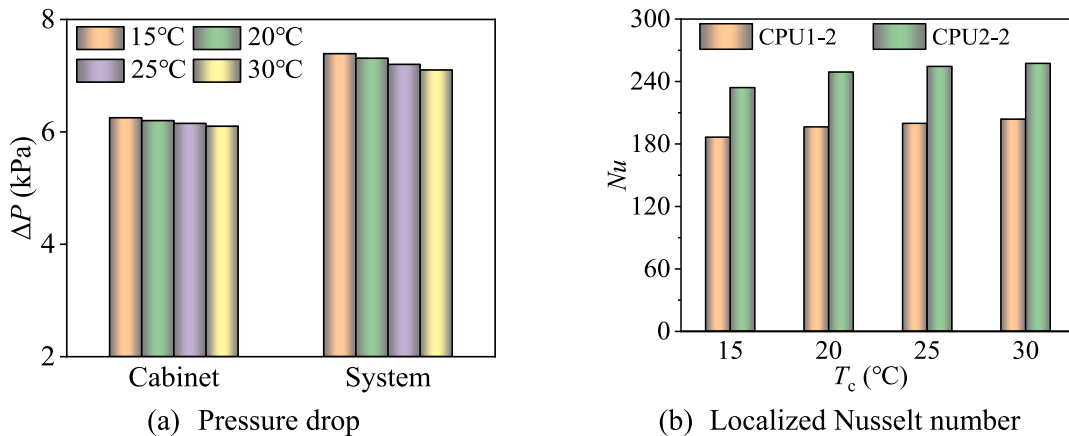


Fig. 13. Effect of water temperature on the forced convection performance.

enhancement; thus, the temperature increase in the demo chip case tends to approach that of the cooling water temperature. Fig. 12(b) further analyzes the EFL temperature uniformity inside the immersion liquid cabinet under different cooling water temperatures. Compared to the case of $T_c = 15^\circ\text{C}$, the temperature variability index of coolants inside the immersion liquid cabinet decreases by 6.05% when $T_c = 30^\circ\text{C}$. This occurs because the increase in cooling water temperature leads to a rise in the EFL temperature, contributing to a reduction in the dynamic viscosity of the EFL and enhanced convective heat transfer, thereby enhancing the EFL temperature uniformity in immersion liquid cabinets.

Fig. 13 evaluates the convective characteristics of the SPILC system using various cooling water temperatures. Fig. 13(a) depicts that rising the cooling water temperature, the pressure drop of the immersion liquid cabinet and the SPILC system continuously reduces. Compared to the case with $T_c = 15^\circ\text{C}$, the pressure drops of the immersion liquid cabinet and the system for the case with $T_c = 30^\circ\text{C}$ decrease by 2.46% and 4.08%, respectively. Interestingly, similar to the case temperature response of demo chips, the decreasing trend of pressure drop gradually weakens as increasing the cooling water temperature. This is because the decrease in EFL viscosity caused by the temperature rise of cooling water, which weakens the flow resistance of coolants and subsequently reduces the system pressure drop. However, the rate of change in dynamic viscosity decreases as the temperature increases, so the decreasing trend of flow resistance caused by the decrease in viscosity gradually becomes less significant. Fig. 13(b) provides the localized Nusselt numbers of CPU2 at various cooling water temperatures. Compared to the case with $T_c = 15^\circ\text{C}$, the localized Nusselt numbers of CPU1-2 and CPU2-2 increase by 9.26% and 9.93%, respectively, when $T_c = 30^\circ\text{C}$. Therefore, it is inferred that although increasing the cooling water temperature results in a rise in the temperature of demo chips, the convective heat transport in the immersion liquid cabinet is strengthened to a certain extent.

3.4.2. Effect of volume flow rate of coolants

To research the influence of different volume flow rates of coolants on the flow and thermal characteristics of SPILC systems, this study quantitatively evaluated the temperature response, flow behaviors, and heat convection capacity in the immersion liquid cabinet. During the experiment, the total power of demo chips in the immersion liquid cabinet is 1.75 kW, and the cooling water temperature is 15°C . Fig. 14 illustrates the temperature response of demo chips and the EFL in the immersion liquid cabinet at different volume flow rates. Fig. 14(a) indicates that the case temperature of demo chips is negatively correlated with the volume flow rate. Moreover, CPU2 is particularly sensitive to

changes in the EFL flow rate because of the influence of the upstream thermal barrier. Quantitative analysis reveals that when the EFL flow rate changes from 2 to 6 L/min, the case temperature of demo chips decreases by a maximum of 10.64°C . However, as the EFL flow rate is further raised to 8 L/min, the maximum decrease in the case temperature of demo chips is only 0.54°C . This phenomenon indicates that increasing the EFL flow rate cannot indefinitely improve the convective heat transport between the EFL and demo chips. There is a critical value for the volume flow rate of the EFL. Beyond this critical value, an increase in the volume flow rate does not dramatically enhance heat transfer intensification. Fig. 14(b) further provides the temperature variability indices of EFLs in the immersion liquid cabinet at various volume flow rates. According to the statistical data, as the volume flow rate rises from 2 to 8 L/min, the temperature variability index of EFLs decreases by 238.29%. Clearly, the increase in the volume flow rate greatly enhances the convective heat transport capacity of EFLs, thereby significantly improving the temperature uniformity of the immersion liquid cabinet. However, this improvement in the EFL temperature uniformity in the immersion liquid cabinet gradually diminishes as the EFL flow rate increases.

Fig. 15 evaluates the convective characteristics of SPILC systems at various volume flow rates. Fig. 15(a) indicates that increasing the EFL flow rate significantly increases the pressure drops of the immersion liquid cabinet and the entire system. Compared to the case with $Q_V = 2$ L/min, the pressure drop of the immersion liquid cabinet and the system for the case with $Q_V = 8$ L/min increase by 414.64% and 635.72%, respectively. This result indicates that although increasing the volume flow rate of EFLs enhances the convective heat transport performance, it also results in a drastic increase in the flow resistance of SPILC systems. Fig. 15(b) further provides the localized Nusselt numbers of CPU2 at different volume flow rates. With the volume flow rate changes from 2 to 8 L/min, the localized Nusselt numbers of CPU1-2 and CPU2-1 increase by 165.63% and 233.56%, respectively. This result further illustrates that increasing the volume flow rate greatly promotes the convective heat transfer of the EFL, enhancing the immersion cooling capacity between the EFL and demo chips.

3.4.3. Control mechanism of SPILC systems

To research the regulatory mechanism of working conditions on SPILC systems, and to provide technical support for the practical operation of SPILC systems. Figs. 16 and 17 illustrate the coupling relationship between different operating conditions and the maximum CPU temperature (i.e., $T_{\max}$), as well as the total electricity consumption (i.e., Q , including power consumption of pumps and chillers). Fig. 16(a) shows that when the volume flow rate of EFLs varies from 2 to 8 L/min

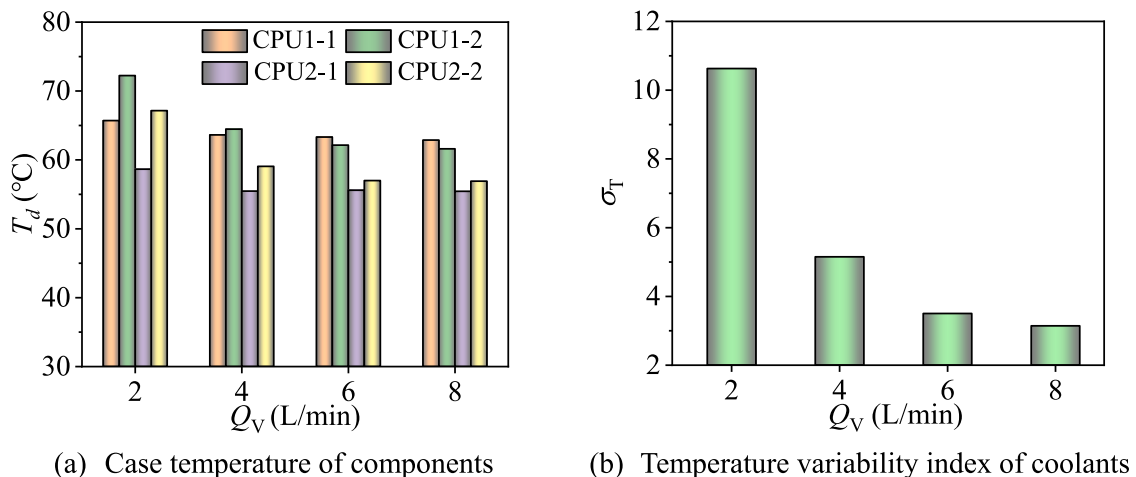


Fig. 14. Effect of volume flow rate on the temperature response.

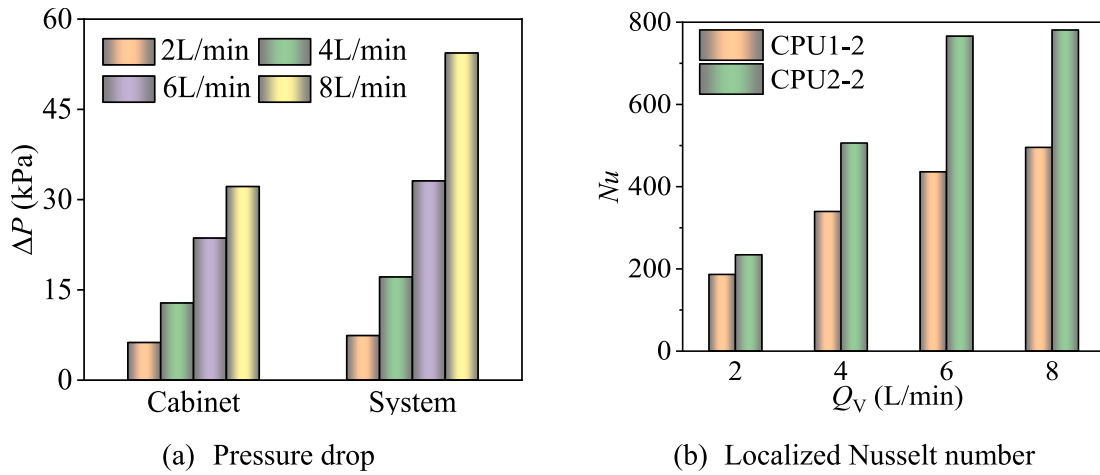


Fig. 15. Effect of volume flow rate on the forced convection performance.

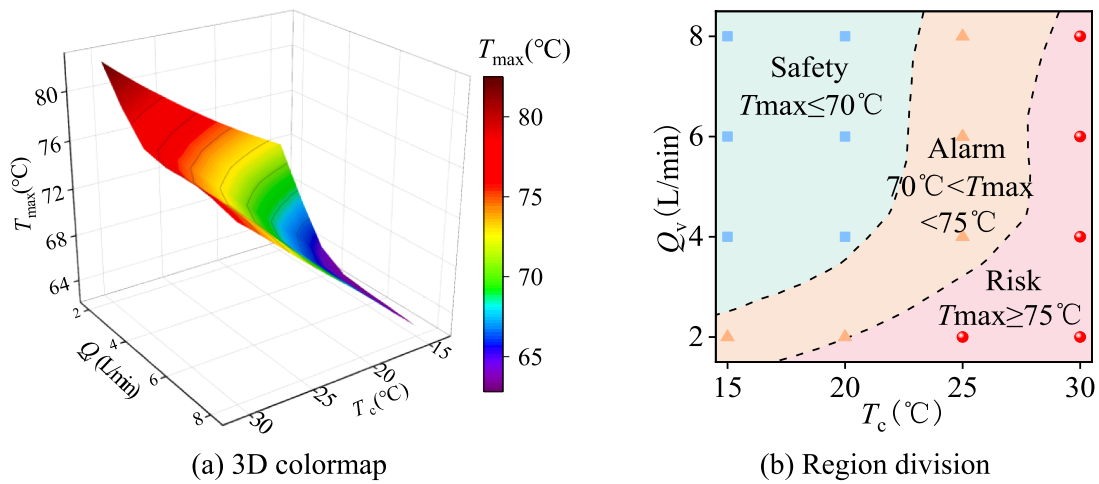


Fig. 16. Maximum CPU temperature in relation to flow rate and water temperature.

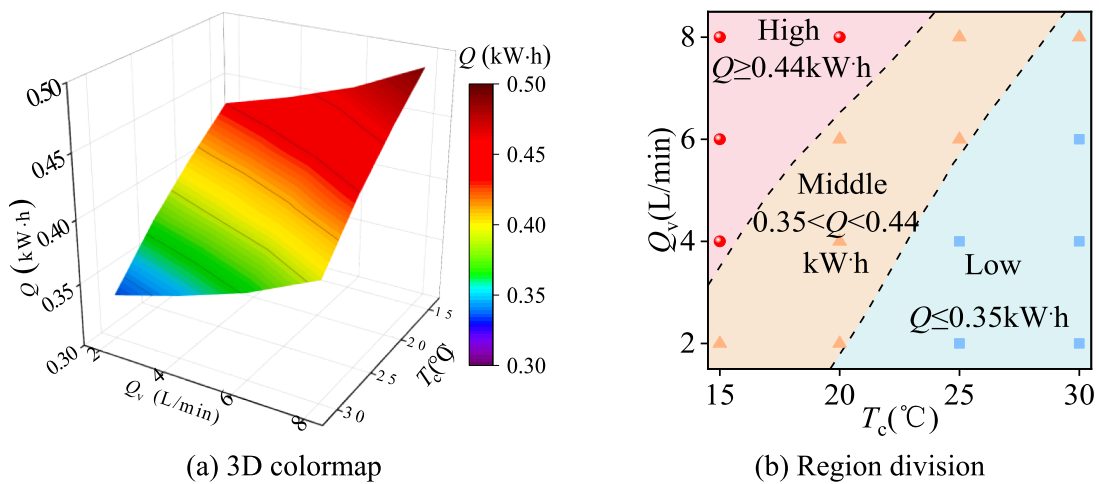


Fig. 17. Total electricity consumption in relation to flow rate and water temperature.

and the cooling water temperature varies from 15 to 30 °C, the maximum CPU temperature in the immersion liquid cabinet fluctuates between 62.8 °C and 82.5 °C. In other words, the CPU temperature undergoes a maximum variation of 19.7 °C under different operating

conditions. Considering the close relationship between CPU temperature and its operational performance, real-time control of operating conditions should be emphasized during the actual operation of SPILC systems. To address this issue, Fig. 16(b) presents a control chart of cooling

water temperature and volume flow rate of EFLs according to the maximum CPU temperature. This control chart shows that the maximum CPU temperature under the coupling effect of different EFL flow rates and cooling water temperatures can be divided into safety, alarm, and risk zones. Careful observation reveals that when $Q_V < 2.48$ L/min and $T_c > 23.4$ °C, the maximum CPU temperature consistently remains outside the safe zone. It can be seen that when the actual SPILC system operates under the same conditions as those in this study, it is necessary to control the coolant flow rate to be greater than 2.48 L/min and the cooling water temperature to be below 23.4 °C. Specifically, when $T_c > 29.1$ °C, the maximum CPU temperature consistently stays in the risk zone, significantly impacting the server's operational performance and even shortening its lifespan. Additionally, the influence of the coolant flow rate on the CPU temperature weakens gradually with an increase in the cooling water temperature. When the cooling water temperature increases from 15 °C to 20 °C, changing the coolant flow rate from 2.5 L/min to 3.6 L/min can bring the maximum CPU temperature into the safe zone. However, when the cooling water temperature further rises to 25 °C, the effect of the coolant flow rate on the maximum CPU temperature becomes almost negligible. In particular, if the CPU temperature does not improve after $Q_V > 8$ L/min, this indicates a failure of the coolant regulation function. Therefore, the EFL volume flow rate and cooling water temperature need to be monitored and adjusted in real-time to ensure the CPU operates normally in the actual operation process.

Fig. 17(a) presents a 3D color-mapped surface plot illustrating the coupling relationship between different operating conditions and the total power consumption of SPILC systems (Ambient temperature is 19 °C). As achieved from the figure, the EFL flow rate has a relatively minor impact on the total electricity consumption. This is because the power of the internal circulation pump in the SPILC system accounts for only 4~23.6% of the chiller power consumption. Considering that the electricity consumption of the chiller is negatively dependent on the temperature of the cooling water, the lower the water temperature, the higher the total electricity consumption of SPILC systems. Fig. 17(b) depicts a control chart of the total electricity consumption of SPILC systems based on the EFL flow rate and cooling water temperature. For quantitative analysis, this study divides the power consumption of SPILC systems into low, medium, and high power consumption zones using PUE values of 1.23 and 1.26 as critical thresholds. By adjusting the cooling water temperature when $Q_V > 3.16$ L/min, the energy consumption of SPILC systems can transition between the low, medium, and high consumption zones. Similarly, when 19.5 °C $< T_c < 24.1$ °C, adjusting the coolant flow rate can facilitate the transition of the SPILC system's energy consumption between the low, medium, and high zones. Furthermore, only when $T_c > 19.54$ °C can the energy consumption of SPILC systems enter the low consumption zone. This phenomenon further indicates that the energy consumption of a chiller is significantly greater than that of the internal circulation pump. Therefore, in the actual temperature control process of SPILC systems, priority should be given to adjusting the internal circulation pump to improve heat transfer performance, while adjustments to the chiller unit's cooling water should only be made when the former has minimal impact.

Based on the above analysis, there are optimal operating conditions for the SPILC system, which ensures that the CPU's maximum temperature falls within a safe range while minimizing the overall electricity consumption of the SPILC system. Moreover, the operation control of the SPILC system should prioritize the CPU temperature as the main objective, with the total system electricity consumption as a secondary objective. When the CPU's maximum temperature is high, the first step is to regulate the EFL flow rate to ensure that the CPU temperature is outside the critical and dangerous zones. If adjusting the EFL flow rate does not improve the CPU temperature, then the cooling water temperature should be adjusted to bring the CPU temperature into the safe zone. In addition, while ensuring CPU safety, efforts ought to be made to raise the cooling water temperature in order to achieve energy-saving

goals.

4. Conclusions

To address the high energy consumption and heat dissipation challenges introduced by the integration, scaling, and high density of data centers, this study focused on the experimental investigation of SPILC systems. By constructing an experimental platform for SPILC cooling, the influence of the EFL flow direction in the immersion liquid cabinet on the SPILC performance was explored. Thus, the inherent relationships between the thermal properties of the EFL and the thermal transport ability, flow behavior, and energy efficiency of a SPILC system were investigated. Furthermore, an in-depth investigation of the influence mechanism of working conditions on the SPILC performance was conducted, and the control mechanism for the SPILC performance under different operating conditions was proposed. In conclusion, the research in this paper leads to the following conclusions.

- (1) With all other conditions kept consistent, adopting the anti-gravity flow direction in the immersion liquid cabinet maximizes the overall performance of SPILC systems. Compared to the case with a pro-gravity flow, the anti-gravity flow reduces the CPU case temperature and thermal resistance by 33.8% and 55.6%, respectively, while increasing the local Nusselt numbers of CPU1-2 and CPU2-2 by 121.2% and 157.8%, respectively. However, the pro-gravity flow direction improves the temperature uniformity in the immersion liquid cabinet by 325% and reduces the system pressure drop by 9.8%. To maintain a consistent maximum CPU temperature, the pump power of the SPILC system decreases by 60.8% under the anti-gravity flow direction, resulting in a 1.4% decrease in the system's PUE.
- (2) The thermal properties of EFLs, especially the dynamic viscosity, play an important role in the SPILC performance. A lower dynamic viscosity helps to weaken the boundary layer on the surface of demo chips and enhances the immersion cooling efficiency. Compared to the other two EFL options, when the immersion liquid cabinet adopts EFL-3 with lower dynamic viscosity, the maximum decrease in CPU case temperature and thermal resistance is 9.3% and 10.5%, respectively. Additionally, the maximum reductions in the system pressure drop and PUE are 6.4% and 0.4%, respectively.
- (3) Increasing the cooling water temperature may increase the demo chip temperature, improve the temperature uniformity of EFLs and reduce the system pressure drop. Additionally, increasing the volume flow rate of EFLs lowers the case temperature of demo chips and improves the temperature uniformity and local Nusselt numbers in the immersion liquid cabinet. However, it also leads to an increase in pressure loss in the SPILC system.
- (4) The operation control of SPILC systems should prioritize CPU temperature as the main objective, with total system power consumption as a secondary goal. Compared to cooling water temperature, the influence of EFL volume flow rate on the heat transfer capacity and energy consumption of SPILC systems is relatively minor. To achieve the lowest energy consumption while maintaining temperature control, the SPILC system should first adjust the EFL volume flow rate. Only after the thermal benefits of regulating the EFL volume flow rate are not significant (i.e., when $Q_V \geq 8$ L/min), should adjustments be made to the cooling water temperature.

CRedit authorship contribution statement

Yongping Huang: Writing – original draft, Methodology, Formal analysis, Data curation. **Bin Liu:** Resources. **Shijie Xu:** Writing – original draft, Data curation. **Chujin Bao:** Validation, Resources. **Yangfan Zhong:** Validation, Resources. **Chengbin Zhang:** Writing – review &

editing, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Nomenclatures

A : surface area of a CPU heat sink
 c_p : specific heat capacity
 d_{min} : minimum value of directly measured parameters
 l : feature length of the CPU heat sink
 F : functional relationship
 n : number of measurement points
 Nu : localized Nusselt number
 p : pressure
 P : power
 Q : power consumption
 Q_v : volume flow rate
 R_{th} : equivalent thermal resistance
 T : temperature
 T_c : cooling water temperature
 T_d : case temperature of a demo chip
 T_{max} : maximum CPU temperature
 T_i : temperature measurement point
 T_{in} : inlet temperature of coolants

Greek symbols

δ_d : directly measured parameter
 δ_i : indirectly measured parameter
 Δp : pressure drop
 ΔT : average temperature difference
 λ : thermal conductivity
 μ : dynamic viscosity
 ρ : density
 σ_p : measurement accuracy of directly measured parameters
 σ_T : temperature variability index

Subscripts

chiller: chiller
IT: demo chip
pump: gear pump

Abbreviations

CPU: central processing unit
DIMM: dual-inline-memory-module
EFL: electronic fluorinated liquid
HDD: hard disk drive
PCB: printed circuit board
PCH: platform controller hub

PSU: power supply unit
PUE: power usage effectiveness

SPILC: single-phase immersion liquid cooling
TPILC: two-phase immersion liquid cooling